

Atomic Structure

Protons, Neutrons & Electrons

A complete lesson guide with diagrams, examples and key facts

What is an Atom? Everything around us is made of tiny particles called **atoms**. They are the **building blocks of all matter** — from the air we breathe to the water we drink to the food we eat! Atoms are incredibly small — millions fit on a full stop.

Part 1 — The Three Particles Inside Every Atom

PROTON ⊕

Positive (+) charge

Lives in the **nucleus**

Determines the **element**

Count = atomic number

Example: Hydrogen has 1 proton — that is what makes it hydrogen!

NEUTRON ■

No charge (neutral)

Also in the **nucleus**

Keeps nucleus stable

Makes **isotopes** possible

Without neutrons the nucleus would fly apart!

ELECTRON ■

Negative (–) charge

Orbits in **electron shells**

Balances proton charge

Determines bonding

Electrons zip around so fast we cannot see them exactly!

Part 2 — How Is the Atom Arranged?

The atom has two main regions: the **nucleus** at the very centre, and the **electron shells** surrounding it.

Region	Contents	Key Facts
Nucleus	Protons + Neutrons	Very dense; tiny compared to the whole atom
Electron Shells	Electrons only	Invisible rings; electrons orbit at high speed

Empty Space	Nothing!	Over 99.9% of an atom is empty space!
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■ **Solar System Analogy:** Think of the nucleus as the **Sun** and electrons as **planets** zooming around it at incredible speed!

Part 3 — Atomic Number & Mass Number

Number	Formula	What it tells us	Example (Carbon)
Atomic Number	= Protons only	Identifies the element	Carbon = 6 protons → atomic no. 6
Mass Number	= Protons + Neutrons	Tells us the atom's mass	6p + 6n = mass no. 12
Neutrons	= Mass – Atomic no.	Calculated from the above	12 – 6 = 6 neutrons

Part 4 — Isotopes

Atoms of the **same element** can have **different numbers of neutrons**. These are called **isotopes**. They still have the same atomic number (same element), but different mass numbers.

Isotope	Protons	Neutrons	Mass Number	Still Carbon?
Carbon-12	6	6	12	Yes
Carbon-13	6	7	13	Yes
Carbon-14	6	8	14	Yes

Part 5 — Fun Facts About Atoms!

■ Mostly Empty!

If an atom were the size of a football stadium, the nucleus would be a tiny pea in the centre!

■ Lightning Fast!

Electrons move so fast around the nucleus that we can never pinpoint exactly where they are!

■ Super Tiny!

There are more atoms in a grain of sand than grains of sand on all the world's beaches!

Part 6 — Practice Questions

	Question	Answer
Q1	Which particle has a positive charge?	Proton
Q2	Where do electrons orbit?	In electron shells around the nucleus
Q3	What is the atomic number of Carbon (6 protons, 6 neutrons)?	6 (number of protons)
Q4	What is the mass number of Oxygen (8 protons, 8 neutrons)?	16 (8 + 8)
Q5	What are atoms with the same element but different neutron counts called?	Isotopes
Q6	Neutrons have what type of charge?	No charge (neutral)

Key Takeaways

⊕	Protons	Positive charge, in nucleus, determine the element (atomic number)
■	Neutrons	No charge, in nucleus, stabilise the nucleus, create isotopes
■	Electrons	Negative charge, orbit in shells, balance proton charge
■	Atomic Number	= number of protons — unique to each element
■	Mass Number	= protons + neutrons — tells us the atom's mass



Isotopes

Same element, different neutron counts, different mass numbers

Keep exploring the world of science!

Atoms make up everything — from the air you breathe to the stars in the sky.